

Does the missing-data method change the answer?

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Research Question and Motivation

Question

In a two-year health survey panel with real attrition i.e. MEPS, how much does the choice of missing-data method change a national estimate of healthcare expenditure and does the choice matter more when the missing data is ignorable (MAR) than when it is non-ignorable (MNAR)?

Motivation

ADEMP Table

Element	My study
Aims	Quantify how missing-data method choice affects mean healthcare expenditure estimate under different data dropout handling mechanisms.
Data-Generating Mechanism	Each replicate draws a bootstrap sample of MEPS truth set; ampute() at 30% under MAR/MCAR/MNAR
Estimand	Mean year 2 total expenditure in truth subset
Methods	Complete-case, mean imputation, MI-PMM, delta-adjusted MI
Performance Measures	Bias, empirical SE, 95% coverage, each with MCSE

Required Limitations

Using MEPS HC-217 (Panel 23, 2018-2019), to study an ordinary dropout mechanism rather than a dataset that muddles the attrition with pandemic-attributed dropout. Cleaner, pre-pandemic, simpler story for my study.

There is not real attrition in the original data file. So my real-world complete case vs mean imputation comparison is comparing methods on data with very little genuine missingness and validates choosing this study design.

Health expenditure from this survey is extremely right-skewed. A mean under 30% missingness is sensitive to a few high-cost respondents, and coverage results should be interpreted with this in mind.

The truth subset being compared against is a selected, healthier-than-average population, so this truth is internal to the survey population, not the national truth.

The simulation runs unweighted.

Results



